written reports, which also provides data triangulation for our analysis.

3. RESEARCH QUESTIONS

This study investigates how student-groups engage in design and science thinking in the given engineering design (ED) task. Though our initial focus was to identify the design–science gap, we transitioned to thinking in terms of design–science connection as the research evolved. We sought to characterize student-groups' design and science thinking and investigate the emerging themes. By analyzing how student-groups' thinking evolved during a multi-week task, we highlight the dynamic interplay between design and science thinking. Additionally, we examined how the laboratory activities, provided as 'scaffolds' through the multi-week task, influenced student-groups' thinking, with the goal of informing more effective educational strategies.

The specific research questions (RQs) are:

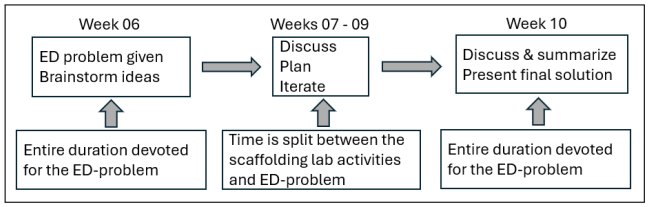
RQ1: *In the context of the prescribed engineering design (ED)-based laboratory task, in what ways can we characterize student-groups' 'design thinking' and 'science thinking', and in what ways do the connections between design and science unfold?*

RQ2: *In what ways do the scaffolds influence student-groups' design and science thinking, and students' progression towards a solution?*

4. METHODOLOGY

A. Context of Study

FIG. 1: Timeline of the ED-based task following the ED Process Model [13, 63, 64]. Each laboratory session lasted 110 minutes. Refer Tables 2 and 3 for details.



This study took place in a large-enrollment, first-semester, calculus-based, undergraduate physics course at a large U.S. Midwestern land-grant university. A significant reform in this course over the last five years has been integrating ED into the laboratory component of the course. In the 2021-22 academic year, the course had an enrollment of about 1000 students in the Fall semester and 1200 students in the Spring semester.

> "Pristine natural habitats of endangered species such as the gorillas in the Congo River basin are becoming increasingly rare. Today, these habitats and the endangered species that inhabit them need to be not only protected but even sustained by humans. As a member of a team of engineers volunteering for a non-profit organization, you are asked to design a system that can launch a payload of food to an island in the Congo River and land it safely for the gorillas. Each payload is about 50 kg, and it must be delivered to a habitat area located on an island in the Congo River that is about 150 m away from the riverbank. To avoid contributing to global warming, the client wants you to use a means that would minimize the carbon footprint of the delivery. Furthermore, the client also wants to ensure that the habitat remains pristine, so that neither humans nor a robotic machine must disturb the flora and fauna of the habitat while delivering the food."

FIG. 2: Engineering Design problem statement provided to the students. Images taken from Wikipedia [65, 66] were also provided.

Students in the course include about 25% women, 10% underrepresented minorities, and about 10% international students. The current study is situated in the Fall of 2021.

A vast majority (over 80%) of the students in this course aspired to be future engineers, while the remaining were science majors. Engineering majors are concurrently enrolled in a first-year engineering course focused on ED, which creates a unified engineering design experience for these students spanning multiple courses. Non-engineering majors are provided tutorials on ED prior to starting the ED problem.

The course adopts a principle-based approach [67], with the content divided into three units, each focused on fundamental physics principles: momentum, energy, and angular momentum. Common threads include a focus on systems thinking, modeling, and making assumptions and approximations. The Momentum Principle [67, ch.2] relates the change in momentum ($\Delta\vec{p}$) of a system to the net force ($\vec{F}_{\text{net}}$) acting on it, multiplied by the interaction duration (Δt). The time interval must be small enough for the net force to remain nearly constant, i.e., $\Delta\vec{p} = \vec{F}_{\text{net}} \cdot \Delta t$. The Energy Principle [67, ch.6] states that the change in energy of the system (ΔE_{system}) plus the change in energy of the surroundings ($\Delta E_{\text{surroundings}}$) equals zero, i.e., $\Delta E_{\text{system}} + \Delta E_{\text{surroundings}} = 0$. The Angular Momentum Principle [67, ch.11] relates a change in angular momentum ($\Delta\vec{L}$) of a system to the net torque ($\vec{\tau}_{\text{net}}$) acting on it, with the interaction duration (Δt), i.e., $\Delta\vec{L} = \vec{\tau}_{\text{net}} \cdot \Delta t$.

The weekly schedule includes two 50-minute lectures, one 110-minute laboratory, and one 50-minute recitation focused on problem-solving. The laboratory segment spanned 13 ses-

sions, with this study focusing on weeks 06 to 10. The laboratory schedule may be seen in Tables 1 and 2.

TABLE 1: Schedule of labs for the semester. This study focuses on ED problem II, occurring in weeks 06-10, hereafter referred to as 'ED-problem'.

Weeks	Lab activities based on	Design Challenge
01 – 05	Momentum Principle	ED problem I
06 – 10	Energy Principle	ED problem II
11 – 13	Angular Momentum Principle	ED problem III

B. Participants

The participants of this study were undergraduate students in two laboratory sections for which the lead author served as the Graduate Teaching Assistant (GTA). From a total of 28 teams, each consisting of 2 or 3 students, data from 14 teams was selected based on our earlier studies [13, 14]. The lead author's position as the TA for the course allowed us to select 14 teams that were representative of the variability of student experiences, thus addressing issues of saturation, under-representation, and over-representation [68].

C. ED problem

Seven essential characteristics of engineering design, as outlined by Capobianco *et al.* [33], guided the development of our ED problem: client-driven and goal-oriented, authentic context, constraints, use of familiar materials, resources, and tools, solution is an artifact or process, multiplicity of solutions, and teamwork.

Students worked in teams to first identify the overall context of the problem, then generated possible ideas or solutions using what they knew about the problem as well as using relevant physics knowledge. Even as the teams executed the scaffolded laboratory activities, in parallel, they developed a plan, applied their scientific knowledge, exchanged ideas with other teams, received feedback from the GTA, made iterations, refined their solution approaches, and presented their final solution in the week 10 lab report. The ED-process model [13, 63, 64] which guided this study is presented in Figure 1.

Although the ED problem (see Figure 2) [65, 66] was situated in the laboratory component of the course, the scaffolding was not confined to the lab alone. Expansive Framing [37] was used to integrate and scaffold learning experiences into the course, including prompting students during lectures to reflect on the relevance of physics concepts to the ED problem and ensuring that weekly recitation problems were related to the same challenge.

Most of the lab sessions, to facilitate the integration of science in ED, were comprised of two interwoven parts designed to aid student groups' progress towards their solutions. The first part involved a traditional hands-on inquiry-based experiment using PASCO equipment [69] and sensors, with data collection and analysis performed using PASCO Capstone™ software [70]. The second part included computational activities using PhET [71] and VPython [72], which is an extension of the Python programming language particularly useful for visualizing physical simulations.

Weeks 06 and 10 were entirely made available for students to engage with the ED problem, while in weeks 07 - 09 the class time was roughly equally divided between lab activities and the ED task. In week 06, they engaged in problem scoping and solution generation (brainstorming). In weeks 07-09, they completed inquiry-based activities related to the ED problem, using hands-on equipment, VPython code, and PhET simulations to help build conceptual and computational models for: (i) launching a payload along an inclined ramp using a spring-loaded device (week 07), (ii) dropping a coffee filter or firing a projectile under the influence of air drag (week 08), and (iii) bouncing a falling object off a hard surface with a certain coefficient of restitution (week 09). Each week, groups were asked to revisit the ED problem and audio record their group discussions in response to a set of prompts presented to them. Students presented their solutions in a combined group report in week 10. The details may be seen in Tables 2 and 3.

As is standard practice in most laboratory sessions, the lead author provided a short introduction to the students about the laboratory activities and the deliverables for the week. TAs are expected to circulate around the class, answering any questions students may have. The lead author was assisted by an undergraduate teaching assistant.

D. Data Collection

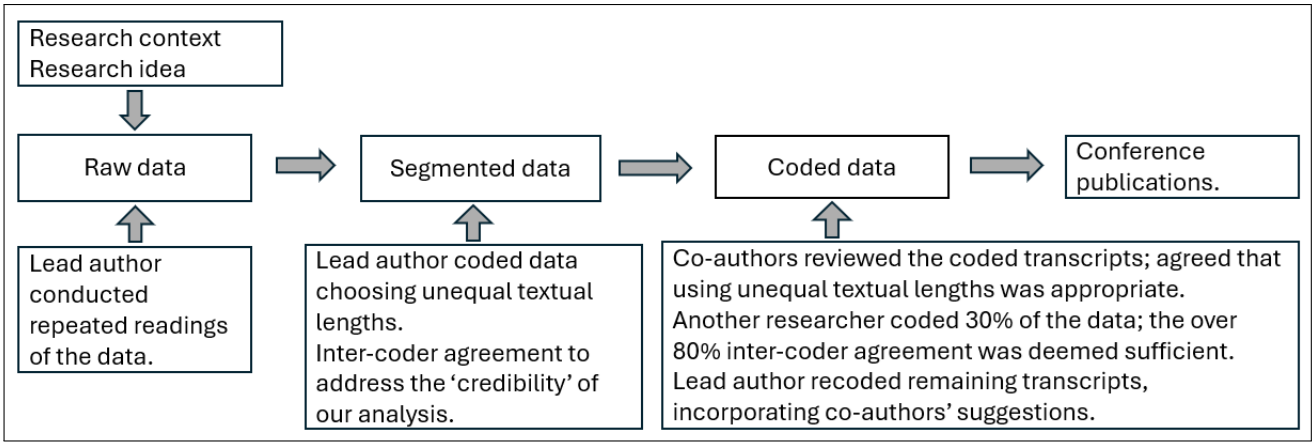
In each lab, from week 06 to week 09, each team of students was asked to audio-record their discussions using the mobile phone of one of the team members. Students were provided a specific set of prompts (see Table 3) to guide their team discussion. We expected that most student-groups would adopt a launch mechanism (we realized later that we may have unintentionally preempted an approach) for delivering the payload. The simulation, hands-on activities, and prompts were all designed with this assumption in mind. However, students were explicitly informed that they could pursue any solution approach.

One team, among the 14 selected, did not submit a written report in the final week. This left us with 56 audio recordings (14 per week) and 13 written reports, providing a substantial volume of data for our qualitative study. Teams were renumbered from Team-01 to Team-14 and will be referred to by these designations for the remainder of this paper.

TABLE 2: Schedule of lab activities, pertaining to the ED-task, provided as scaffolds within the laboratory in weeks 06 to 10. Audio-recordings of group discussions and written reports submitted by the students form the data for this study.

Week	Lab activities (Scaffolds)	Deliverables
06	Students presented with the ED problem. Student-groups brainstorm ideas for possible solution approaches.	Written lab report + an audio recording of the group discussion in response to prompts - Problem scoping and solution generation.
07	Hands-on task: Exploring springs: Force and Energy - Motivating a launch mechanism (Initial stage). VPython simulation-1.	Written lab report for the hands-on task. Audio recording of group discussions in response to ED problem related prompts – Discussing the initial stage.
08	Hands-on task: Exploring Drag Force: Motivating use of a parachute (Middle stage). Dropping a coffee filter or firing a projectile under the influence of air drag. VPython simulation-2. PhET simulation.	Written lab report for the hands-on task. Audio recording of group discussions in response to ED problem related prompts – Discussing the middle stage
09	Hands-on task: Exploring Coefficient of Restitution (COR). Motivating soft- landing the payload (Final stage). VPython simulation-3.	Written lab report for the hands-on task. Audio recording of group discussions in response to ED problem related prompts – Discussing the final stage.
10	Final review and submission of ED problem solutions. VPython simulation-4.	Summarize the several iterations of the group's solution and submit a written lab report containing group's proposed solution to the ED problem.

FIG. 3: Coding Process - Stage 1. Transcripts were coded by textual segmentation. Refer Section 4 E (i), Tables 4 and 5.



E. Data Analysis

Student-groups recorded discussions in the laboratory or corridors, and the resulting background noise made it extremely challenging to uniquely associate conversation segments with individual speakers. Moreover, the final report in week 10 was a group effort. These factors, along with the fact that students worked in groups in all labs, influenced our decision to choose the *"grain size"* or *"unit of analysis"* [15, p.10] to be that of a group. Each recording, ranging from 3 to 5 minutes, was manually transcribed and coded by the lead author. This traditional 'pen and paper' *"codus operandi"* [73, p.22] allowed for direct engagement with the data, offering valuable insights through close interaction with the material. The text length of the transcripts ranged from about 400 to 800 words each. The lead author meticulously took handwritten notes while reviewing the raw data multiple times, documenting thoughts and decisions in Word documents and Excel spreadsheets. He drew inspiration from the words of B. O'Dwyer:

> *"Whatever process of analysis you use, there is no substitute for knowing your data intimately"* [74, p.404].

To address *"reliability"* and for *"ensuring rigor"* in our qualitative inquiry, we were guided by Morse *et al.* [75, p.14]. The lead author engaged in periodic *"peer debriefing"* [75, p.14] sessions with the co-authors, not only to review and refine the coding schema, but also to develop an iterative *"constructive"* [75, p.15] procedure for the study. The data analysis evolved through two stages.

TABLE 3: ED task-related prompts presented to students from weeks 06 to 10. During weeks 06 to 09, responses were captured through audio recordings of group-discussions, while week 10's response was submitted as a written report.

Week	Prompts
06	Brainstorm multiple potential solutions to this problem. Focus on key ideas, principles/concepts involved, and limitations. Complete solutions will be developed in subsequent weeks.
07	Reflect on how today's lab activity contributes to your understanding of the ED problem. In what ways did you approach the initial stage (the launch) of food delivery to the gorillas, and how might you approach it differently now? Did your ideas change, and what aspects of the lab made you think differently about the design challenge?
08	Reflect on how today's lab activity contributes to your understanding of the ED problem. In what ways did you approach the middle stage (after the launch) of food delivery to the gorillas, and how might you approach it differently now? Did your ideas change, and what aspects of the lab made you think differently about the design challenge?
09	Reflect on how today's lab activity contributes to your understanding of the ED problem. In what ways did you approach the final stage (the landing) of food delivery to the gorillas, and how might you approach it differently now? Did your ideas change, and what aspects of the lab made you think differently about the design challenge?
10	Restate the problem. Include the design context, identifying stakeholders, mention the criteria and constraints, and outline the significance of the problem. Describe the iterations in the problem-solving approach, list relevant physics principles and concepts, outline your assumptions and approximations., and state the limitations in your design. Present a sketch for your design.

(i). Stage 1 Coding

In Stage 1 (see Figure 3), our initial efforts were focused on investigating the design–science gap [13, 14]. In this stage we looked into students' 'ways of thinking' [47, 76], and the basis of students' statements. The codes with detailed descriptors and examples are presented in Tables 4 and 5. In both cases, textual segments of varying lengths were chosen as coding units. This flexible approach was adopted after careful consideration of factors such as the complexity of student discussions and the unique conversation styles within each group [15, 77]. The selection of coding units was guided by factors such as communication styles, thematic shifts, speaker transitions (this being the most challenging to decipher from the audio), contextual completeness, and, above all, the nature of our research questions. According to Elliot [78, p.2856], *"there is no simple answer"* to the choice of *"the chunk of data"* to be coded, and *"it depends on the study and your aims within it"*. Furthermore, fixed text lengths might overlook the richness and diversity of these discussions, as some groups engaged in dynamic exchanges while others spoke more linearly. It is important to point out that multiple codes (code co-occurrence) [79], [78, p.2853] can be applied to a single segment. To adjust for the variable lengths of the transcripts and the use of unequal textual lengths, we calculated fractional coding frequencies for each team and averaged them across all groups.

An important consideration in our analysis was whether to include error bars in the bar charts. In an earlier study [14], we included error bars because our arguments relied on code frequencies, necessitating the establishment of statistical significance for differences in those frequencies. However, for the current study, the qualitative nature of our work posed challenges in meaningfully defining error, particularly given that the researcher functions *"as an instrument"* [80–83] in qualitative research. Standard practices for error bars, such as using standard deviation, standard error, or confidence intervals, are not directly applicable. Although custom error bars could be considered, they would still require a clearly defined notion of coding error, which remains ambiguous, perhaps even questionable, in qualitative research.

Despite extensive searches, we found no clear guidance in the literature on the use of error bars for bar charts in qualitative research. Given our focus on the interpretive aspects of qualitative data rather than the statistical significance of code percentages, we decided not to include error bars in Figures 7 and 8. Out of curiosity, we queried AI platforms such as Perplexity AI, ChatGPT, and MS Copilot regarding the appropriateness of error bars for such data. Their responses were notably consistent, stating that *"Error bars on bar charts of code frequencies from qualitative data are generally inappropriate unless quantifiable measures of variability are included"* [84–86]. While insights from AI platforms may not be definitive, we believe this perspective may be of interest to the research community. These considerations further supported our decision to exclude error bars in this study.

The code percentages [78, p.2853] from Stage 1 coding are majorly used only in two of the Findings & Discussions sections, namely: Sections 5 B and 5 C.

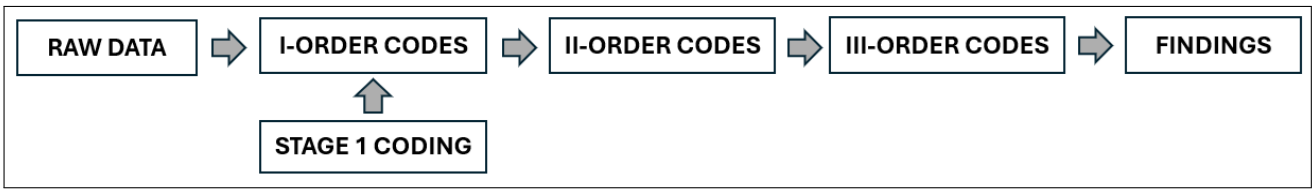
(ii). Stage 2 Coding

Initially guided by the first stage of coding, we aimed to enhance the level of detail. To address concerns about the unequal lengths of text segments in stage one coding, we inductively developed a more comprehensive and multi-layered coding scheme, grounded in the data. For data triangulation [17, p.37], we used multiple data sources—audio recordings and written reports. The coding process for this stage is

TABLE 4: Codes for 'Ways of Thinking', code descriptions, and example quotes.
DST - Design Thinking; SCT - Science Concepts; MAT - Mathematical Thinking; MER - Metacognitive Reflection.

Code	Code Descriptor	Illustrative Quote	Code Justification
DST	Define the problem, identify constraints, brainstorm multiple solutions, iterate, select the best solution, consider design issues / aspects, prototype the solution, test, communicate.	*"an electric drone which could carry enough weight. It could fly a payload over and drop off and leave without really interacting with anything. The problem could be the charge of the drone. Yeah we don't know how long the distance is and it might be difficult to charge it in the middle of the Congo. It might be hard to find a place to take off and to land. Lifting 50 kg will be hard for a drone to lift. It'd have to be one heck of an electric drone."* (Team-05; Week 06)	Identifies a design constraint - Can the drone carry 50 kg? Addresses design issues - How to charge the drone? Addresses a design criteria - Not to disturb the habitat.
SCT	Organization, cause and effect, systems, scale – absolute and relative, models, change and rate of change. In our context, it may be expected that the students refer to concepts of projectile motion, drag force among other things.	*"I would say that this spring system is probably not the best idea because at least with what we've found with our data the displacement of the spring with the amount of mass we were using was not enough to get a whole lot of force. And with the velocity they're trying to achieve for the actual problem itself this does not seem a very likely solution unless we had a very long launching system."* (Team-07; Week 07)	Referring to physics concepts - Considering if it may be practical for a spring to have large enough compression and spring constant to generate enough speed for launch. Energy conversion implied. There is some element of DST in terms of what type of a spring may be required for the design.
MAT	It may refer to a scientific statement in terms of a relation among several variables and constants. Proportional reasoning is acceptable. Units analysis. Use of explicit expression / equation.	*"I am just estimating here … and shooting it that way so you get more altitude … with more altitude and the shape of the object considered more air resistance and further slowing down."* (Team-03; Week 08)	Proportional reasoning - How drag force increases with distance of fall. There is probably some SCT in the form of an indirect reference to the formula for drag force.
MER	Reflect on their ideas / recall previous notions and comment / reflect and modify / critically review.	*"so after this lab what we've realized is that either this new discovery of the coefficient of restitution … will either hurt launching an object or help it because we could either destroy the environment by not bouncing or we can retain some energy and bounce it multiple times instead of letting out all the energy on the environment."* (Team-06; Week 09)	Reflecting on their learning post the lab activity with bouncing balls. Also coded for - DST - Paying attention to the design criteria that habitat's integrity should not be compromised. SCT - Thinking about the energy loss due to impact.

FIG. 4: Coding Process - Stage 2. Data Reduction inspired by the Gioia Chart [20, p.21]. Written reports included for 'Triangulation'. Refer to Appendices A and B.



depicted in Figures 4 and 5. Separate coding schemata developed for the transcripts and written reports may be seen in Appendices A and B (or Tables 6 and 7).

For data reduction [79, p.13], we were guided by the data structure model proposed by Gioia *et al.* [20, p.21], which is illustrated in Figure 4. This detailed, multi-layered approach helped to enrich our qualitative analysis, providing deeper insights into our research questions, particularly concerning the 'design–science connection'.

The coding schema underwent several revisions, and only the final version is presented in this study. Two co-authors independently assisted with the coding after being briefed on the project's goals. An inter-coder reliability (ICR) exceeding 80 percent was deemed sufficient in both stages of the coding process, a decision taken after much debate and research. As to what may be an appropriate percentage of ICR agreement, we observe that *"there is little consensus on how to approach reliability in qualitative research"* [87, p.72:18]. McDonald *et al.* even caution that, for some studies, an ICR test may be *"potentially harmful"* [87, p.72:19], a view echoed by O'Connor and Joffe who argue that an ICR test may not be appropriate for every qualitative study and should not be considered a *"magic bullet"* [88, p.11] to automatically ensure reliability. We were committed to documenting our analysis and while Stage 1 coding was useful, its insufficiency led us to engage in more nuanced Stage 2 coding. However, we were wary that relying too heavily on code percentages could undermine the qualitative nature of our study. These concerns led us to prioritize 'thick description' [17–19, 89, 90] to enhance the trustworthiness of our findings. The minor differences in coding were not deemed significant enough to affect the results, particularly since 'thick description' allows *"the reader to enter the research context"*[18, p.26],[17, p.38]. We also questioned whether an ICR was necessary when using thick description for our analysis. Although we did not find satisfactory answers, we chose to retain coding for two key reasons: (i) we began with coding and had detailed results, and (ii) coding helped us verify consistency between themes and thick descriptions [88, p.2]. The detailed coding process provided a valuable audit trail that guided our discussions and decisions when finalizing the themes and descriptions. Based on our experience, we believe that coding is a valuable first step for beginners in qualitative research who wish to employ thick description. It provides an organized framework for structuring data and ensures that rich, detailed narratives emerge from the analysis.

(iii). Guiding Frameworks

Given our context, research goals, research questions, the nature of our data, and the type of analysis we conducted, we found useful guidance in a variety of theoretical or conceptual frameworks in different stages of our analysis.

Following the principle *"letting the data speak to the researcher"* [91, p.8], we strove to let the data guide our analysis. Our approach involved flexible notions of design and science thinking, and the iterative development of a coding scheme from substantial data sources (group discussions and written reports). Given the inductive nature of our analysis and the evolving focus of our study, we received - only for the purposes of data analysis - some guidance from the grounded theory method approach outlined by Urquhart [91].

Our coding process followed the Gioia Framework [20], while the presentation of our findings was guided by the MIRACLE framework for Thick Description. In our thematic analysis [16], [92, p.8], we employ thick description (with the authors' interpretation) and direct student quotes (to capture the informants' voices). The MIRACLE framework's seven key criteria [89, p.7, 8, and 9] informed the articulation of our results: Meaningful (clear articulation of participants' thinking), Interpretative (balanced consideration of participant perspectives and researchers' interpretations), Rational (alignment of methods, data, and findings within the research context), Authentic (genuine representation of participants' thinking), Contextualized (interpretations drawn from the data, with researchers' biases made transparent), Linked (cohesive connections between methods, research questions, and findings), and Emic (detailed, context-centered account).

In examining the interplay between student groups' design and science thinking, Mestre's Transfer of Learning framework [41] offered valuable guidance. Furthermore, the concept of *"distributed scaffolding"*, as articulated by Kolodner and Puntambekar [9], proved instrumental helping us analyze the impact of scaffolds on students' thinking.

Most importantly, since our data is grounded in a physics laboratory, our analysis and interpretation were strongly guided by our disciplinary understanding of Newtonian Mechanics. While students occasionally drew on ideas from other areas of physics, their science thinking predominantly centered around concepts of motion, force, and energy, framed within the conceptual framework of Newtonian Mechanics.

The co-authors reviewed the themes and narrative descriptions, and the lead author incorporated their feedback to ensure consistency with the coding. Extensive discussions were held on the most effective ways to present the qualitative findings. Figure 6 depicts the process flow diagram from Data to Findings.

5. FINDINGS & DISCUSSION

In this section, we present our findings on how the selected 14 student groups navigated the prescribed engineering design (ED) task. We focus on the emergent themes in students' design and science thinking, and how the design-science connections unfolded in their work, addressing RQ1. To address RQ2, we examine how the scaffolds—particularly the lab activities—may have influenced students' thinking. These findings are drawn from both stages 1 (see Figure 3) and 2 (see Figure 5) of our coding process, with the latter

TABLE 5: Codes for 'Behind the Words: The Basis of Student Statements', code descriptions, and example quotes. HT - Hands-on Task; VP - VPython and PhET Simulations; PK - Physics Knowledge; GK - General Knowledge.

Code	Code Descriptor	Illustrative Quote	Code Justification
HT	Students make references to the hands-on laboratory experiments using PASCO hardware and software. Could be the current lab / the previous, including those in the weeks 01–05.	*"This was basically Hooke's law. So we're going to be using that. Apply to the catapult with the spring force and the stretch. We've to calculate the force to the stretch ratio and hopefully do the calculations."* (Team-05; Week 07)	Directly referring to the hands-on laboratory activity. This was also coded for PK, probably gained from coursework.
VP	Students make references to the PhET simulations and / or VPython simulations. Since it would be hard to decipher which simulation activity the students may be referring to, it was decided to club them together.	*"Also how inconsistent our actual data is from simulations and stuff... because our actual data from capstone is very different than the Python simulation ... yeah that's true we had a technical error yeah ... so calculating drag force .. like by looking at a video link manually is much more difficult than ... digitally ....obviously. "* (Team-01; Week 08)	Making references to the simulation activities in the lab. Also coded for HT - Referring to the hands-on lab activities with data collection via PASCO software.
PK	Statements made based on their physics knowledge gained via coursework in school / university, and from textbooks.	*"Do we have a theoretical idea if we use a parachute about how much the speed will reduce by or do we have to do the calculations for that. We probably have to do a timed parachute release."* (Team-03; Week 06)	Referring to calculation methods possibly based on coursework. The reference to 'timed parachute release' probably reflects GK.
GK	Statements based on common knowledge and everyday experience.	*"Do not know how to say it in English... But it's a place where you can jump. Oh, a trampoline yes yes instead of a parachute so it feels safe and eco-friendly or maybe like an airbag, something like you know how like you know when people jump out of a building they use huge air sacks ... when they practice snowboarding.. you do not need to worry about bouncing off. It kind of like absorbs it."* (Team-02; Week 06)	This is during the brainstorming session of week 06. Statements based on experience or common knowledge.

stage playing a more prominent role. It is important to note that our discussion does not rely solely on coding; we majorly employ thick description to convey the researchers' perspectives, and incorporate participants' voices through carefully and thoughtfully chosen direct quotations. As Ponterotto rightly notes:

> *"A thickly described Discussions section of a qualitative report successfully merges the participants' lived experiences with the researcher's interpretations of these experiences..."* [19, p.547].

In stage 1, we coded only the transcripts, for which the coding schemata may be seen in Tables 4 and 5. Figures 7 and 8 visually represent the variation of code percentages (for all groups taken together) across weeks 06 to 09. In stage 2, as previously mentioned, we coded both transcripts and written reports. The coding schemata for this stage are detailed in Appendices A and B, with Figures 9 and 10 offering visual representations of the coding results.

In Section 5 A, we outline the variety of solution approaches considered by the student-groups. Section 5 B explores the variation in students' design, science, mathematical, and reflective thinking through the multi-week task. Section 5 C examines the factors that may have influenced students' thinking during the task and provides an overarching view of how the provided scaffolds shaped their thought processes. These sections offer partial insights into both of our research questions.

In Sections 5 D through 5 I, we delve deeper towards addressing our research questions. Section 5 D has a particular focus on students' design thinking, while Section 5 E investigates their physics thinking. Section 5 F explores the blurred boundaries between the often-perceived dichotomy of design and science. The importance of Mathematical Thinking in science and engineering can never be discounted, and this aspect is discussed in detail in Section 5 G. We conclude by offering our perspectives on the notion of the 'design-science gap' in Section 5 I, ultimately emphasizing the need for educational objectives to enhance the connection between design and science. Although these sections primarily address RQ1,

the reader will notice how the discussion, quite unavoidably, also intersects with RQ2.

Our response to RQ2 is interwoven throughout most sections, with Sections 5 A, 5 C, 5 D (iii), 5 E (iii), 5 F (i), and 5 F (iv) addressing it more explicitly.

Since this was a multi-week activity, we were particularly interested in how students reflected on their design and science thinking throughout the process. Section 5 H addresses this, simultaneously drawing connections to both research questions.

As previously noted, students' thinking emerges in complex ways, and readers will observe that all sections maintain some connection to both research questions. Our analysis is guided by a thematic exploration, without being confined to a single team or a specific set of teams. This study aims to uncover patterns that can inform future interventions, while also providing insights to guide pedagogical decisions, with an awareness of the limits of generalization [93].

A. Variety of Solution Approaches

While our analysis remains largely agnostic to the solution approaches taken by student groups, it would be illustrative to highlight some of the initial solution paths considered by students.

The most popular approach involved launching the load using a catapult or trebuchet, closely followed by the use of drones and ziplines. Other innovative ideas included raft systems, automatic underground tubing systems, amphibious automated guided vehicles, autonomous sky-cranes, boats, tugboat ferry systems, pressure guns, and airplane drops, among others. These diverse approaches reflect the students' extensive general knowledge, creativity, willingness and eagerness to explore out-of-the-box ideas, and interdisciplinary knowledge. It demonstrates their ability to think critically about the design problem, weigh in various science principles, engage deeply with the project, and explore a wide range of technologies and methods. Among the interesting conversation snippets during the brainstorming session (week 06) was from Team-03: *"We could do a composter-cannon. What do you guys think about that? ...Actually what is it? Explain what it is.... Like you'd have a cannon that uses compressed air to launch the payload and you'd have a parachute attached to the payload so that it wouldn't damage any of the trees or wildlife. The parachute would also be biodegradable..."*. The reader would note that this conversation segment also captures a thought-provoking interaction among the group members. The manner in which students seamlessly interweave science - not just physics - concepts into their design is palpable. Interestingly, the idea of a composter-cannon was entirely novel to the lead author before this study, highlighting that there is much we can learn from our students.

Opinions on adopting a launch mechanism varied widely, ranging from mild criticism to outright rejection. Team-07 offered a gentle and balanced view, noting: *"I would say that this spring system is probably not the best idea because at least with what we've found with our data the displacement of the spring with the amount of mass we were using was not enough to get a whole lot of force"*. The reader may also observe that the team's thinking is guided by the scaffolding hands-on activity (see Table 2) in week 07. Conversely, Team-09 in week 07 expressed strong reservations about the launch mechanisms, feeling constrained by the scaffolds. They remarked that insisting on a catapult *"would require a very big and very powerful spring so even though this lab was informative I don't think it's necessarily a good solution to the problem and not necessarily a practical solution.. I mean .. I think they're going to push us towards springs ..I think that's what they're going to give us next. But there are better launching methods"*. This 'feedback' underscores the importance of remaining open to alternative student solution approaches, in addition to highlighting an important lesson for us: we may have prematurely favored a sub-optimal approach. What is particularly notable in these conversation segments is student-groups' motivation to debate and refine their ideas. It is easy to note that the students are thinking in terms of the magnitude of the spring constant while referring to a "very powerful" spring, and about dimensions of their contraption while referring to a "very big" spring. While some teams were uncertain about the best approach, Team-06 clearly identified what to avoid and proceeded with their rocket-based contraption (see Figure 11), which we will address in more detail in the later sections.

The initial brainstorming session facilitated the generation of a wide range of ideas. As educators, we got a sense of the range of solution approaches students explored. It may be noted that the classroom instructions merely asked students "brainstorm" their ideas without any specific guidance on how to brainstorm. We wonder how our data and findings might have differed if we had offered our students a more structured guidance on brainstorming techniques. For our future iterations we intend to make use of the four key rules for effective brainstorming proposed by Kaufman and McCuish [94, p.7, 8]: avoid judgment; prioritize quantity over quality; embrace unconventional ideas; and build upon each other's contributions.

In their written reports, most teams settled on the catapult approach or a variation of it. This could be due the influence of the accompanying scaffolds - hands-on and virtual lab activities (see Table 2) - that may have prompted students to think about the catapult solution. A few of the teams chose alternative solution methods such as drones and ziplines.

We acknowledge that some, or perhaps all, of these solution approaches may not fully meet the task requirements. However, our goal was to remain objective in evaluating the approaches and not be judgemental of students' work. Accordingly, the spirit of our coding schema is to remain respectful to students' choices.

learning science: An exploratory study, Science education **84**, 566 (2000).

[134] A. S. Huffmyer and J. D. Lemus, Graduate ta teaching behaviors impact student achievement in a research-based undergraduate science course, Journal of College Science Teaching **48**, 56 (2019).

[135] L. White, V. Svihla, Y. Chen, T. Hynson, I. Drackert, J. James, C. Saul, and A. Megli, Validating a measure of problem-framing ability to support evidence-based teaching practice, in *Proceedings of the ASEE 126th Annual Conference and Exhibition* (2019).

[136] H. Radder, Science, technology and the science-technology relationship, in *Philosophy of technology and engineering sciences* (Elsevier, 2009) pp. 65–91.

[137] L. P. Davis and S. D. Museus, What is deficit thinking? an analysis of conceptualizations of deficit thinking and implications for scholarly research, NCID Currents **1** (2019).

[138] L. Shiner, "blurred boundaries"? rethinking the concept of craft and its relation to art and design, Philosophy Compass **7**, 230 (2012).

[139] E. Eisner and K. Powell, Art in science?, Curriculum Inquiry **32**, 131 (2002).

[140] J. D. Team, Jupyter (2024), accessed: 2024-12-02.

[141] R. C. Subramaniam, J. W. Morphew, C. M. Rebello, and N. S. Rebello, Presenting a stem ways of thinking framework for engineering design-based physics problems (2024), arXiv:2411.11654 [physics.ed-ph].

APPENDIX A

TABLE 6: Stage 2 Coding. Multi-layered coding schema for coding transcripts (lab 06 - 09). Data Reduction is inspired by the Gioia Chart [20, p.21]. The I-order codes are closest to the raw data (transcripts). Example statements for the codes (not given here for want of space) may be evident either as actual quotes or through descriptions in the 'Findings & Discussion' section. DST - Design Thinking; SCT - Science Thinking; DA - Design Aspects; CC - Criteria and Constraints; ME - Metrics; SH - Stakeholders; PCP - Physics Concepts & Principles; MAT - Mathematical Thinking; ES - Environmental Science; MS - Material Science.

III-order code	**II-order code**	**I-order code**
DST	DA	Economics - Cost
		Mechanisms / Design Detail - no / low mention
		Mechanisms / Design Detail - moderate / high detail
		Ensure safe landing
		Environmentally safe
		Design limitations
		Incorrect / inaccurate statements
	CC	Payload weight
		150 m - distance
		Minimum C-footprint
		Habitat Integrity
	ME	Delivery accuracy
		Payload dimensions
		Carbon footprint
		Habitat Integrity
	SH	Engineers / Human workers
		Flora and Fauna
		Others
SCT	PCP	Thinking about the path; Trajectory calculations.
		Physics principles - low / no detail
		Physics principles - elaborated
		Physics terms / facts / ideas/concepts - low / no detail
		Physics terms / facts/ideas / concepts - elaborated
		Conceptually incorrect / factually incorrect statements.
	MAT	Stated, in low / no detail.
		Presented in some detail.
	ES	Mere mention - direct / indirect.
	MS	Mere mention - direct / indirect.

APPENDIX B

TABLE 7: Stage 2 Coding. Multi-layered coding schema for coding written reports (Lab 10). Data Reduction is inspired by the Gioia Chart [20, p.21]. The I-order codes are closest to the raw data (written reports). Example statements for the codes (not given here for want of space) may be evident either as actual quotes or through descriptions in the 'Findings & Discussion' section.
DST - Design Thinking; SCT - Science Thinking; DA - Design Aspects; CC - Criteria and Constraints; ME - Metrics; SH - Stakeholders; DL-Design Limitations; AA - Assumptions & Approximations; ; FC - Physics Facts & Concepts; ; PP - Physics Principles (Momentum, Energy, Angular Momentum Principles); MAT - Mathematical Thinking; ES - Environmental Science; MS - Material Science; IT - Iterations; SK - Sketch/Diagram.

III-order code	**II-order code**	**I-order code**
DST	SH, CC, ME	Does the written work have anything significantly new/different as compared to the group discussions?
	IT-Level 1	Team states skeletal details. There is very little meaningful iteration. There's nothing new compared to what's seen in the group's prior discussions.
	IT-Level 2	Team states skeletal details. There is some meaningful iteration. There are a few changes the group has effected in the report building on the prior discussions.
	IT-Level 3	Team provides significantly more details, not seen in the prior discussions.
	SK-Level 1	The sketch is skeletal, with barely anything new to the text / earlier discussions.
	SK-Level 2	The sketch has new information not seen in the text / earlier discussions.
	SK-Level 3	The sketch has significant information beyond text / earlier discussions.
	AA-Level 1	Vague statements.(see Section 5 F (iii)
	AA-Level 2	Cursory / generic statements. (see Section 5 F (iii)
	AA-Level 3	Thoughtful statements. (see Section 5 F (iii)
	DL-Level 1	Vague statements.
	DL-Level 2	Cursory / generic statements.
	DL-Level 3	Thoughtful statements.
SCT	FC-Level 1	Mostly a repetition of what was present in the prior discussions.
	FC-Level 2	There were a few new facts, but nothing significant.
	FC-Level 3	Several new ideas were presented, but not elaborated.
	FC-Level 4	Several new ideas were presented and elaborated.
	PP-Level 1	Mostly repetition of what was present in the prior discussions.
	PP-Level 2	Team elaborated on the principles used in prior discussions.
	SK-Level 1	The sketch is skeletal, with barely anything new to the text / earlier discussions.
	SK-Level 2	The sketch has new information not seen in the text / earlier discussions.
	SK-Level 3	The sketch has significant information beyond text / earlier discussions.
	AA-Level 1	Vague statements. (see Section 5 F (iii)
	AA-Level 2	Cursory / generic statements. (see Section 5 F (iii)
	AA-Level 3	Thoughtful statements. (see Section 5 F (iii)
	ES	Nothing new when compared to the discussions.
	MAT-Level 1	No math.
	MAT-Level 2	Some equations / formulas in text.
	MAT-Level 3	Some equations / formulas in sketch.